Zhenlin Zhu

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SUMMARY

Data Scientist with a **PhD** in astrophysics and 6+ years of experience in statistical modeling, machine learning, and large-scale data analysis. Proficient in Python and SQL, with hands-on experience building end-to-end pipelines for data processing, feature engineering, predictive modeling, and evaluation. Experienced in recommendation systems, A/B testing, and deep learning, with additional expertise in LLM fine-tuning and AI agents. Skilled at translating stakeholder needs into analytical pronlems and communicating actionable insights to technical and non-technical teams.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL, Shell
LLM and AI: Continued Pretraining, Instruction Tuning, LangGraph Agents, RAG, NLP
Machine Learning: Deep Learning, Recommendation Systems, Gradient Boosting, Random Forests, Clustering, A/B Testing
MLOps and Cloud: Docker, Jenkins, Airflow, AWS, Alibaba Cloud ECS, PySpark, MLflow, CI/CD

WORKING EXPERIENCE

AI Engineer | *Donglian Tech* Sichuan, China | July 2026 - Sep 2026

- **Domain LLM Training**

Trained a **27B** telecommunications LLM through continued pretraining and instruction fine-tuning; evaluated model performance on the Open Telco AI benchmarks and internal network optimization tasks, achieving **18% improvement** over the base model and stronger performance on real-world network optimization tasks.

- **RCA Agent Development**

Built a LangGraph-based Root Cause Analysis (RCA) agent for network troubleshooting, implemented with tool calling, conditional routing and retry handling; integrated it with the main agent to support multi-turn hypothesis tracking, evidence validation, and traceable conclusions for network troubleshooting.

- **Deployment & CI/CD**

Deployed the RCA agent on Alibaba Cloud ECS and exposed the agent services through MCP and REST APIs; Containerized model and agent services with Docker and implement CI/CD workflows using Jenkins for automated testing, integration, and deployment.

AI4S Data Scientist | *University of California, Los Angeles* Los Angeles, CA | Nov 2024 – Jun 2026

- **Astronomical ETL Pipelines**

Developed scalable SQL and Python pipelines to extract, clean, transform, and load large-scale unstructured astronomical data for downstream analysis and visualization. Applied adaptive-binning and image-sharpening to reveal hidden physical structures.

- **Supervised Learning & Source Classification**

Built an end-to-end feature extraction and multi-class classification pipeline from raw data, training tree-based models (Random Forest, XGBoost, LightGBM) on **5,000+ samples** and achieving **0.93 macro ROC-AUC.**

- **Unsupervised Learning & Clustering**

Applied Gaussian Mixture Models (GMM) to **cluster 8,000+** astronomical sources, using Bayesian Information Criterion to optimize the number of clusters. Analyzed cluster characteristics to identify distinct source population.

- **AI Agent Benchmarking**

Contributed expert-validated scientific coding tasks to **ScienceIDE benchmark**, defining the task setup and executable scientific checks used to evaluate AI agents on correctness beyond code compilation.

Machine Learning Engineer Intern | *Eth Tech* Newark, CA | May 2025 – Aug 2025

- Fine-tuned Salesforce's BLIP imaging captioning model, improving **BLEU-4 scores by 16%** over baseline.
- Built LLM-powered product metadata enrichment with Sentence-Transformer and name entity recognition.
- Developed a cohort retrieval system based on product click-through rates and automated updates with Airflow, improving recommendation freshness.
- Built a two-tower retrieval model with negative sampling; boosted **CTR by 1.2%** in **A/B testing**.
- Implemented incremental training workflows with Airflow for automated model refresh, improving total user actions by **0.6%**.

EDUCATION

Ph.D in Astrophysics *Leiden University* 2019 – 2024
M.S. in Astrophysics *Nanjing University* 2016 – 2019
B.S. in Astronomy *Nanjing University* 2012 – 2016
- 15 peer-reviewed papers (6 as first author), 30+ talks at leading conferences around the world.